SMART COMPLAINT MANAGEMENT SYSTEM

UNIT TEST REPORT

Test execution summary and results

Project detail	Value
Group	8
Course	Software Engineering (UE24CS341A)
Institution	PES University
Version	1.0
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PREPARED BY

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1. Purpose

This report records unit and API-level verification performed for the Smart Complaint Management System (SCMS). It summarizes the latest automated test execution, test scope, outcomes, and verification environment.

2. Test Scope

Account authentication, role checks, and portal-specific access.

Complaint creation, validation, ownership, tracking, status handling, and priority calculation.

Assignment workflows, employee actions, department scoping, reassignment, and evidence validation.

Serializer validation, complaint service rules, audit/history, dashboard data, resolution, closure, and reopening.

Authorization failures, invalid inputs, empty results, and selected transaction rollback scenarios.

3. Test Environment

Component	Recorded configuration
Operating system	Windows / PowerShell
Python	3.13.15 (project virtual environment)
Backend	Django REST Framework
Coverage tool	Coverage.py 7.17.2; branch measurement enabled
Frontend	React / Vite 8.3.4
Frontend commands	npm run lint; npm run build

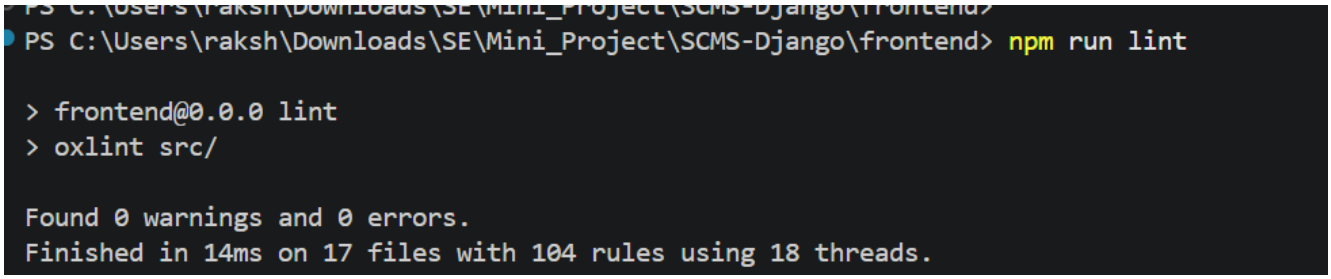


Figure 1: Frontend lint verification with zero warnings and errors.

4. Execution Summary

Metric	Result
Django automated tests executed	103
Passed	103
Failed	0
Overall result	PASS
Frontend lint	0 warnings and 0 errors across 17 files
Frontend production build	Successful; 86 modules transformed
git diff --check	Passed

The latest supplied coverage-instrumented full Django run completed with 103 passing tests and no failures.

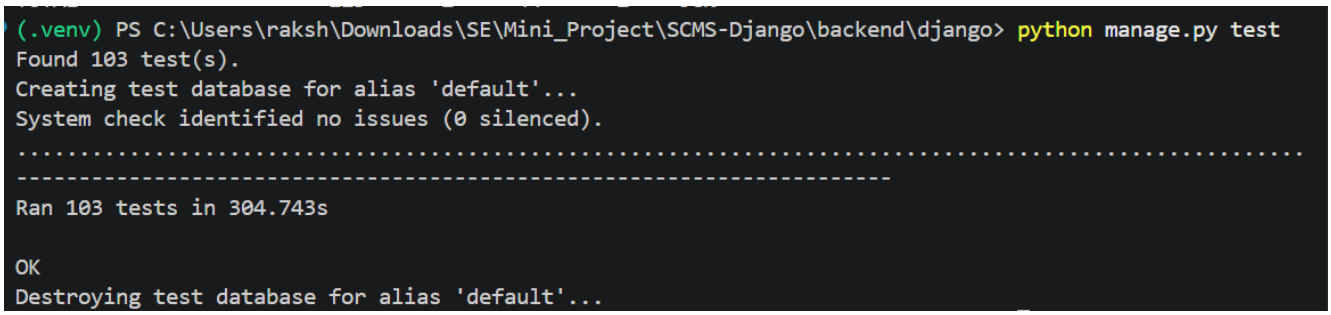


Figure 2: Successful execution of the complete Django automated test suite (103 tests passed).



Figure 3: Successful Vite production build.

5. Representative Test Areas

Area	Representative checks	Result
Authentication & roles	Role-specific login; reject role mismatch	Passed
Complaint tracking	Ownership, status and update visibility	Passed
Assignments	Employee workflow, department scope, evidence authorization	Passed
Priority policy	Thresholds, zero/positive counts, text fallbacks	Passed
Serializers	Required/optional fields, invalid values, attachments	Passed
Complaint services	Creation, transitions, resolution, reopen, audit, rollback	Passed
Audit & dashboard	Role scoping, filters, empty data, profileless admin	Passed

6. Defects Corrected During Testing

Progress API response serialized a stale assignment; it now serializes the updated assignment.

Department Admin without a department profile is denied organization-wide workload access.

Explicit affected count of zero now contributes zero affected-user priority weight.

7. Limitations

This report records automated unit/API-level testing; it does not claim exhaustive production load testing.

No independently measured response-time or throughput benchmark was supplied.

Execution evidence should be retained with the submitted report.

8. Conclusion

The latest reported Django test run completed with 103 passing tests and no failures. Frontend lint and production build checks also passed. The test suite exercises the principal complaint lifecycle, role-based access, assignments, audit, dashboard, validation, and error-handling paths.